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The chosen project is the fourth one, the DBMS comparison, highlighting advantages and disadvantages.

The goal of the project is to compare performances between PostgreSQL, a RDBMS and Neo4j that is a Graph DBMS, in the context of airplane traffic network.

For this purpose I have found three different datasets:

1. Openflight datasets (https://github.com/jpatokal/openflights/tree/master/data)
   Contains 3 principal datasets: airports(airports.dat), airlines(airlines.dat) and routes(routes.dat)
2. A dataset of flights in USA in 2015 on Kaggle (flights.csv)
   (https://www.kaggle.com/datasets/usdot/flight-delays/data?select=flights.csv)
   This contains the entire traffic of USA flights during 2015, for each flight several data are collected, such as date, flight number, tail number, schedule departure time, actual departure time, and more
3. A dataset of flights in USA in 2013 on Kaggle (flights.csv)
   (https://www.kaggle.com/datasets/mahoora00135/flights?resource=download)
   This dataset is similar to the previous, but is slightly less detailed

The idea is to use the Openflight datasets, containing airports, airlines, and routes as the core part. In this way can be conducted different types of analysis, like how many hops there are between an airport and another(here the expectation is that graph DB performs well). As route is intended the connection between an airport and another. The datasets on Kaggle instead, takes flights already done, so contain lot of information for every flight, not only source, dest and company. They also contain day, flight number, departure delay, arrival delay and others. These things could be used to perform another type of analysis, using also aggregation operators, in order to analyse differences between one type and another of DB architecture. (here instead the expectation is that relational DB performs well)

Since two datasets , of 2015 and 2013, have slightly different information the idea is to keep only interesting information that are in both datasets, to maintain data homogeneous.